

An ACLED-Based Knowledge Graph for LLM-Driven Conflict Analysis via GraphRAG

Course Project Development Presentation

Tommaso Arone (896282) Clelia Meloni (895982) Lorenzo Triolo (895541)

Data Semantics – Course Project

June 16, 2026

Presentation Outline

- ➊ Motivation & Research Questions
- ➋ Dataset & Exploratory Analysis
- ➌ Semantic Knowledge Graph Engineering
- ➍ Graph Analytics & GraphRAG Pipeline
- ➎ Query Answering Engine
- ➏ Routing Classification
- ➐ Evaluation, Results & Limitations
- ➑ Conclusions & Future Work

Motivation & Core Idea

The Tabular Limitation

- ACLED provides flat records (one row per conflict event).
- Actor, location, and source relations remain implicit.
- Relationships between actors, locations, and sources are not explicitly represented, making cross-entity analysis more difficult.

The Semantic Solution

- Transform records into a Knowledge Graph with explicit nodes and typed edges.
- Use SPARQL for deterministic and reproducible querying.
- Use the Knowledge Graph to provide structured context to LLMs through GraphRAG.

Overall Goal

Demonstrate how transitioning from a flat table to semantic modelling changes what can be queried, retrieved, and reasoned over in global conflict analysis.

Research Questions

- ① **Visibility:** To what extent does a Knowledge Graph representation expose geopolitical relationships that are difficult to capture in a flat CSV format?
- ② **Queryability:** Can SPARQL querying support deterministic, structured, and reproducible analysis of complex multi-actor armed conflicts?
- ③ **Reasoning:** Can the graph structure provide useful context for GraphRAG-based reasoning about conflict dynamics?

- Source: Armed Conflict Location & Event Data Project (ACLED).
- Access: REST API with OAuth 2.0 authentication.
- Scope: Middle East region, 15 countries.
- Time span: 2015–2023.
- Dataset size: 91,893 events.
- **GitHub Source Code**

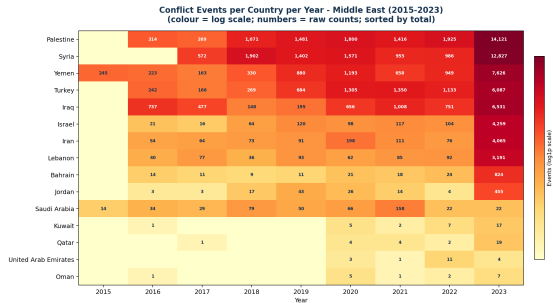
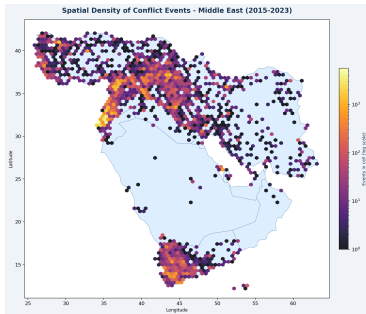
Key Attributes

event id, event date, event type, sub-event type, actor1, actor2, country, coordinates, sources, fatalities, narrative notes.

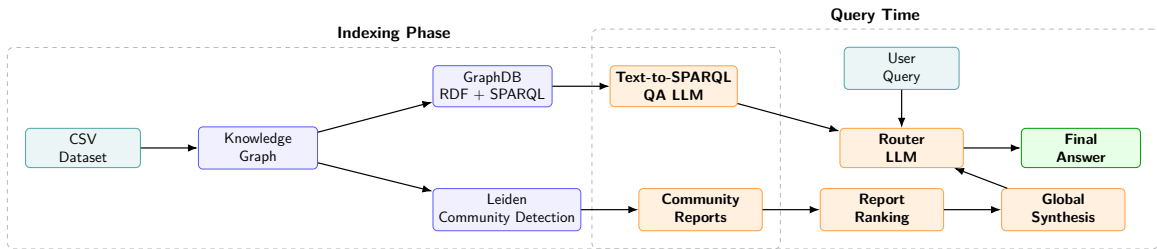
Exploratory Data Analysis: Spatial & Temporal

Architecture Drivers

- **Spatial (Hex-bin):** The hex-bin map shows that conflicts are highly localized. This supports modelling relationships at the actor and event level instead of considering only country labels.
- **Temporal (Heatmap):** It reveals strong fluctuations in conflict activity over the years. We therefore apply temporal sub-sampling when building GraphRAG reports to keep the input manageable for the LLM.



Main Architecture



Local Search: GraphDB → Text-to-SPARQL QA

Global Search: Leiden Communities → Community Reports → Global Synthesis

LLM Model Selection & Rationale

Design Principle

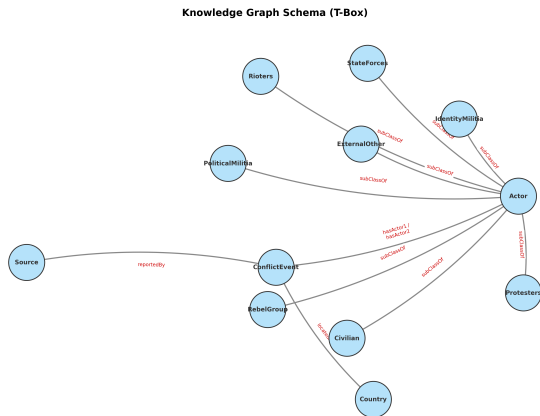
Two backends: **Groq** (Llama) for low-latency structured workflows; **Gemini** for extensive context windows or iterative semantic tasks. Models are assigned based on specific task requirements.

Component	Model	API	Rationale
Community Report Builder	gemini-2.5-flash	Gemini	Generates structured Markdown reports from clustered graph entities. Requires optimal context capacity to ingest multi-year conflict data and compile summaries efficiently.
Router	llama-3.1-8b-instant	Groq	Binary intent classification (SPARQL vs. GraphRAG). A smaller Llama model on Groq keeps routing latency sub-second.
SPARQL Generator	llama-3.3-70b-versatile	Groq	Translating NL to formal logic requires high parameter scale. The 70B model minimizes syntax errors and enforces the OWL schema exactly.
GraphRAG Scorer	gemini-2.5-flash-lite	Gemini	Ranks community report summaries (0–10). A lightweight model is efficient, fast, and sufficient for basic scoring tasks.
Answer Synthesizer	gemini-3.5-flash	Gemini	Synthesizes comprehensive community data into a narrative response. Gemini's large context window handles multi-year text without truncation.
LLM-as-Judge	gemini-2.5-flash-lite	Gemini	Evaluates faithfulness and relevance. Leverage semantic evaluation to verify that claims are strictly grounded in retrieved contexts.

Design Summary: Groq / Llama handles high-speed, structured logic (generation, routing). Gemini handles large text ingestion and evaluation (scoring, synthesis).

Ontology Design

- Main classes: ConflictEvent, Actor, Country, Source.
- Actor subclasses follow the ACLED inter-type taxonomy.
- Object properties: hasActor1, hasActor2, locatedIn, reportedBy.
- Data properties use XSD types for temporal and numerical querying.



Triple Store Configuration

Local RDF store providing a SPARQL endpoint and OWL ontology hosting.

- **Environment:** Isolated Docker container.
- **Endpoint:** localhost:7200 (SPARQL endpoint).
- **Data:** OWL Ontology + `acled_kg.ttl`.

Architectural Advantages

- **Subclass Inference:** Supports OWL subclass hierarchies; in practice, explicit UNION patterns are used to keep queries efficient.
- **Semantic Direction:** Ontology constraints (domain and range definitions) help maintaining consistent relationships between entities.
- **HTTP Integration:** Python modules query GraphDB through its HTTP SPARQL endpoint and retrieve JSON results.

Leiden Algorithm

Objective Identify densely connected groups of actors within the conflict network, such that each community represents a coherent conflict ecosystem.

Network Definition: $G = (V, E)$

- V : conflict actors extracted from ACLED
- E : co-occurrence relationships between actors
- $w(i, j)$: number of conflict events shared by actors i and j

Algorithm Workflow

Local Moving $\rightarrow$ Refinement $\rightarrow$ Aggregation $\rightarrow$ Iterate

- **Local Moving:** actors are assigned to communities that improve the partition quality.
- **Refinement:** communities are internally re-evaluated to ensure strong internal connectivity.
- **Aggregation:** refined communities become super-nodes and the process is repeated on a higher-level graph.

Output A hierarchy of well-connected communities maximizing network modularity and preserving meaningful conflict structures.

Topological Analysis & Community Rebuilding

1. Initial Leiden Detection

- Actor co-occurrence graph extracted via SPARQL.
- Algorithm identifies **306 fine-grained communities**.
- *Quality*: High Modularity (0.72) and Low Conductance (0.01).

2. Semantic Meta-Graph Merging

- **Problem**: 306 reports fracture the narrative & exceed context limits.
- **Solution**: Merge using a similarity score (spatial, temporal, semantic overlap).
- **Result**: Strategically compressed into **17 target GraphRAG communities** with High Modularity (0.713) and Low Conductance (0.015).

3. Community Quality Metrics

Modularity

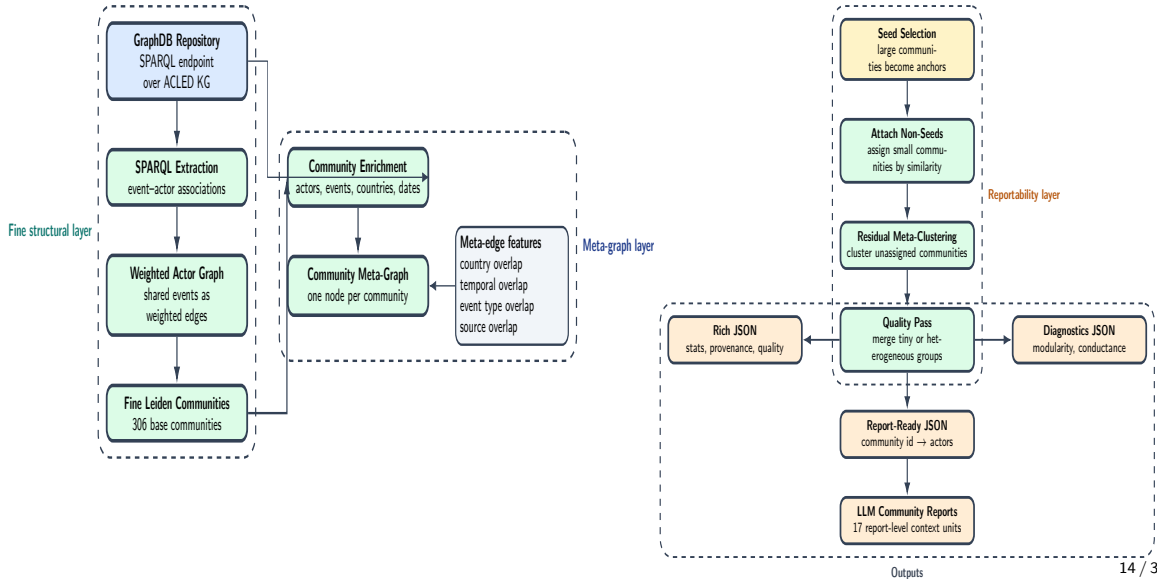
$$Q = \frac{1}{2m} \sum_c \left(e_c - \frac{K_c^2}{2m} \right)$$

Conductance

$$\phi(S) = \frac{\text{cut}(S, \bar{S})}{\min(\text{vol}(S), \text{vol}(\bar{S}))}$$

- c : community; m : total edge weight; e_c : internal edge weight of c ; K_c : total degree of nodes in c .
- S : node subset/community; $\bar{S}$: complement; $\text{cut}(S, \bar{S})$: edges crossing outside S ; $\text{vol}(S)$: sum of degrees in S .
- High Q and low ϕ indicate dense, well-separated communities.

GraphRAG Community Rebuild Workflow



The 17 Final GraphRAG Communities

Community	Actors	Period	Dominant country	Total events
graphrag_community_002	250	2016-01-02–2023-12-31	Syria	27,050
graphrag_community_004	221	2015-01-03–2023-12-31	Yemen	19,396
graphrag_community_010	154	2015-04-02–2023-12-06	Syria	278
graphrag_community_007	121	2016-01-01–2023-12-31	Iraq	11,965
graphrag_community_013	114	2015-03-15–2023-12-22	Syria	161
graphrag_community_001	94	2016-01-06–2023-12-31	Palestine	30,324
graphrag_community_006	78	2016-01-02–2023-12-31	Turkey	11,554
graphrag_community_011	77	2016-03-13–2023-12-14	Syria	122
graphrag_community_008	41	2015-03-05–2023-12-31	Iran	6,339
graphrag_community_009	37	2016-01-01–2023-12-31	Bahrain	1,822
graphrag_community_012	33	2016-12-24–2023-12-21	Lebanon	131
graphrag_community_014	31	2023-01-02–2023-12-07	Yemen	84
graphrag_community_005	27	2015-04-17–2023-12-31	Iraq	14,416
graphrag_community_003	18	2016-01-04–2023-12-31	Palestine	23,213
graphrag_community_015	14	2023-01-14–2023-12-12	Syria	72
graphrag_community_016	11	2017-12-26–2023-12-06	Yemen	48
graphrag_community_017	2	2023-11-07–2023-11-07	Saudi Arabia	2

Observation

Despite the algorithmic merging process, the resulting 17 macro-communities perfectly recover distinct national conflict theatres and temporally bounded escalations. Sometimes the dominant country could be repeated, mainly due to different event type (battles, protests...).

Community Report Generation

Temporal Sub-sampling

- **Challenge:** Macro-communities span up to 30,000 events over 8 years.
- **Solution:** Chronologically stratified sampling extracts the **top 50 most lethal events per year**, guaranteeing context-window compliance without losing critical escalation turning points.

Algorithmic Severity Profiling

Before LLM generation, we computed three custom weighted metrics (log-smoothed): *Historical Severity*, *Current Severity (2023)*, and *Escalation Risk*.

LLM Prompt Engineering

Model: Google Gemini 2.5 Flash

The prompt enforces a strict Markdown structure grounding the LLM entirely on the injected events:

- ① YAML Frontmatter (injecting exact computed severity scores)
- ② Executive Summary
- ③ Primary Actors & Dynamics
- ④ Temporal Evolution (2015-2023)
- ⑤ Severity & Strategic Assessment

Severity Metrics

Historical severity

$$I_{\text{hist}} = |E| + 5 \sum_{e \in E} F(e)$$

$$S_{\text{hist}} = \min \left(100, \frac{\log(1 + I_{\text{hist}})}{\log(1 + 50000)} \cdot 100 \right)$$

- Uses all historical events.
- Fatalities have higher weight.
- Captures long-term severity.

Current severity

$$I_{\text{curr}} = |E_{2023}| + 5 \sum_{e \in E_{2023}} F(e)$$

Current severity

$$S_{\text{curr}} = \min \left(100, \frac{\log(1 + I_{\text{curr}})}{\log(1 + 10000)} \cdot 100 \right)$$

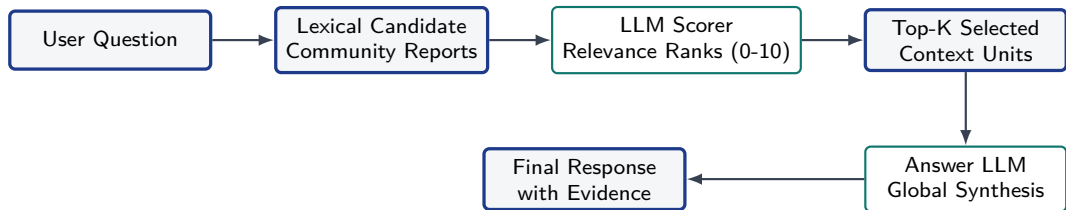
- Uses only events from 2023.
- Measures recent intensity.
- Can differ from historical severity.

Escalation risk

$$R_{\text{esc}} = \left(\frac{N_{2023} - N_{2022}}{N_{2023} + N_{2022}} + 1 \right) \cdot 50$$

$$R_{\text{esc}} = 50 \quad \text{if } N_{2023} + N_{2022} = 0$$

Narrative Engine: GraphRAG Search Pipeline



1. Two-Stage Retrieval (Gemini2.5 Flash-Light)

- **Lexical Pass:** Flags candidate community reports matching entity mentions.
- **LLM Scorer:** Filters and ranks contexts (0–10) to prune irrelevant regional data.

2. Grounded Synthesis (Gemini 3.5 Flash)

- **Model Rationale:** Massive context window handles complex longitudinal data without token truncation.
- **Synthesis:** Leverages pre-computed YAML metrics to evaluate macro-theatres cleanly.

GraphRAG in Action — Live Example

User Query (Global Search)

"Tell me about Palestine."

Retrieval Step

Lexical pass matches "Palestine" to candidates.

LLM Scorer (Gemini 2.5 Flash-Light) ranks 0–10:

community_001	10/10
community_003	8/10
community_005	5/10
community_006	3/10
community_007	3/10

Top-3 reports injected as context into the **Synthesis LLM** (Gemini 3.5 Flash).

Generated Answer (excerpt)

The conflict in Palestine is part of the larger *Levant Conflict Cluster*, involving Israeli Military and Police Forces and Palestinian Armed Groups (Hamas, Palestinian Islamic Jihad) [community_001].

Severity metrics are at their maximum: Historical Severity = 100.0, Current Severity = 100.0, Escalation Risk = 98.6 [community_001].

The conflict dynamics include property destruction, weapons-related incidents, and periodic intense armed clashes with significant humanitarian consequences [community_001, community_003].

[Answer grounded exclusively on injected community reports — no parametric hallucination.]

Query Answering Engine

Design Choices (Llama 3.3 70B)

- **Schema injection:** full TBox (namespaces, class hierarchy, typed properties) in system prompt anchors the LLM to valid predicates.
- **Style guardrails:** explicit UNION for bidirectional actor lookup; LIMIT 20; no ambiguous ASK forms.
- **Deterministic execution:** HTTP POST to GraphDB → raw JSON → NL formatting with thousand separators.

Natural Language Input

"How many times was Israel involved in the events?"

System Answer

Israel was involved in the events **32,757** times.

Pattern: UNION of location-based and actor-name-based lookup ensures complete coverage regardless of how Israel appears in the triple.

Generated SPARQL Query

```
PREFIX conf: <http://data-semantics-2526.org/
          acled/ontology#>

SELECT (SUM(?c) AS ?total) WHERE {
  {
    SELECT (COUNT(DISTINCT ?ev) AS ?c) WHERE {
      ?ev a conf:ConflictEvent ;
          conf:locatedIn ?country .
      ?country conf:countryName "Israel" .
    }
  }
  UNION
  {
    SELECT (COUNT(DISTINCT ?ev) AS ?c) WHERE {
      ?ev a conf:ConflictEvent .
      { ?ev conf:hasActor1 ?actor }
      UNION
      { ?ev conf:hasActor2 ?actor }
      ?actor conf:actorName ?actorName .
      FILTER(CONTAINS(LCASE(?actorName),
                      "israel"))
    }
  }
}
```

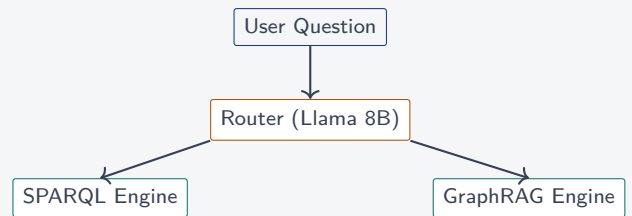
Pipeline Architecture: Hybrid Q&A Router

The Orchestration Layer

A central gateway classifies user intent in real-time to dispatch the question to the optimal execution engine.

- **Intent Classification (Llama 3.1 8B):** Evaluates whether the question requires deterministic metrics or synthesis.
- **Routing Criteria:**
 - *SPARQL Route:* Precise filters, dates, coordinates, or mathematical aggregates.
 - *GraphRAG Route:* Global patterns, entity relations, or thematic summaries.

The Routing Hub



Router in Action — Classification Examples

Llama 3.1 8B classifies each query as requiring **deterministic metric retrieval** or **semantic synthesis**.

[1] Narrative → GRAPHRAG

"Tell me about the escalation of the Israel–Palestine conflict."

Why GraphRAG: No specific metric requested. Requires integrating multi-year actor dynamics and escalation patterns across communities [001, 003, 012].

Triggers: evolution, dynamics, alliances, summary, describe, "tell me about... ", role of

[2] Factoid → SPARQL

"How many times was Palestine offended by Israel?"

Why SPARQL: Explicit quantitative aggregate (*how many times*) with named actors and directional filter. GraphDB returns exact count: **21,220 events**.

Triggers: count, how many, which date, coordinates, total fatalities, list of, exact figure

Dual-Track Evaluation Framework

The system is evaluated across two parallel tracks to assess both deterministic accuracy and semantic reasoning capabilities.

1. Factoid QA (Text-to-SPARQL)

- Validated against a *Golden Dataset* of calculated quantitative questions.
- **Metrics:** Pipeline Router accuracy, Query syntactic validity, and Exact Match rate.

2. Narrative QA (GraphRAG)

- RAGAS-Inspired LLM-as-Judge Evaluation.
- **Metrics:** *Faithfulness* (grounding on retrieved reports) and *Answer Relevance*.

Evaluation Metrics — Definitions

Metric	Track	Definition
Router Accuracy	SPARQL	<code>decide_route(q)</code> matches the expected route (golden label)
Valid SPARQL Rate	SPARQL	Query is well-formed <i>and</i> executes on GraphDB without error
Exact Match (EM)	SPARQL	Normalised golden value found in the NL answer (numeric or substring match)
Faithfulness (0–10)	GraphRAG	Answer grounded in retrieved community reports, no hallucinations
Answer Relevance (0–10)	GraphRAG	Answer actually addresses the question asked

GraphRAG Judging Protocol

Single holistic call to `gemini-2.5-flash-lite (T=0)` $\rightarrow$ {relevance, faithfulness, rationale}. *RAGAS-inspired, not canonical*: no statement-level decomposition / NLI verification.

Results: Factoid Query Answering

- The Text-to-SPARQL module was tested on queries requiring strict aggregations, temporal filters, and topological connections (e.g., actor network interactions).

Quantitative Metrics (SPARQL Track)

Metric	Score
Pipeline Router Accuracy (Task Classification)	100%
Valid SPARQL Generation Rate	100%
Exact Match Accuracy (Golden Dataset)	95%

Key Insight: Schema-aware prompting combined with few-shot examples effectively mitigates syntax hallucinations, ensuring reliable extraction of exact answers directly from the graph.

Observation: The Test set is based on only 20 questions given the API free rate limit.

Results: GraphRAG

- Narrative queries requiring longitudinal analysis (e.g., "Evolution of Syrian alliances") were evaluated by an LLM Judge on a 0–10 scale.

LLM-as-a-Judge Evaluation (RAGAS inspired)

Metric	ACLED GraphRAG
Faithfulness (Absence of Hallucinations)	9.30 / 10
Answer Relevance (Alignment to Prompt)	9.10 / 10

Discussion on LLM-as-a-Judge

LLM-as-a-Judge provides a scalable and low-cost alternative to manual evaluation, allowing rapid assessment of Faithfulness and Answer Relevance across many queries. However, the methodology is not free from bias, since the evaluator is itself a language model. Consequently, the reported scores should be interpreted as indicative rather than absolute measures of quality.

Observation: The Test set is based on only 10 questions given the API free rate limit.

Problems and Limitations: Temporal Query

User Query

Which actors increased their activity the most after 2020?

GraphRAG Response

Based on the selected community reports, it is difficult to pinpoint which actors increased their activity the most after 2020 with certainty, as the reports are aggregated and do not provide specific data on the level of activity of each actor over time.

The reports highlight the involvement of actors such as the Israeli Military and Police Forces, Hamas Movement, Palestinian Islamic Jihad (PIJ), Hezbollah, ISIL, Military Forces of Syria, Houthi Movement, Saudi-led Coalition, and others.

Given the complexity of these conflicts and the lack of specific time-series data, it is challenging to determine which actors have increased their activity the most after 2020.

Discussion

The response is not hallucinated: all mentioned actors are grounded in the retrieved community reports. However, it fails to answer the user question because the task requires temporal aggregation and ranking rather than narrative synthesis. This failure highlights a routing limitation: temporal trend questions may appear descriptive but actually require symbolic reasoning over the Knowledge Graph.

Problems and Limitations: Temporal Query

Text-to-SPARQL Query

```
SELECT ?actorName ?eventsBefore2021 ?eventsAfter2020
      (?eventsAfter2020 - ?eventsBefore2021 AS ?increase)
WHERE {
  { SELECT ?actorName (COUNT(DISTINCT ?eventBefore) AS ?
    eventsBefore2021)
    WHERE {
      ?eventBefore a conf:ConflictEvent ; conf:eventDate ?dateBefore .
      { ?eventBefore conf:hasActor1 ?actor }
      UNION { ?eventBefore conf:hasActor2 ?actor }
      ?actor conf:actorName ?actorName .
      FILTER(?dateBefore < "2021-01-01"^^xsd:date)
    }
    GROUP BY ?actorName
  }
  { SELECT ?actorName (COUNT(DISTINCT ?eventAfter) AS ?eventsAfter2020)
    WHERE {
      ?eventAfter a conf:ConflictEvent ; conf:eventDate ?dateAfter .
      { ?eventAfter conf:hasActor1 ?actor }
      UNION { ?eventAfter conf:hasActor2 ?actor }
      ?actor conf:actorName ?actorName .
      FILTER(?dateAfter >= "2021-01-01"^^xsd:date)
    }
    GROUP BY ?actorName
  }
}
ORDER BY DESC(?increase)
LIMIT 20
```

SPARQL Result

Actor	Increase
Protesters (Turkey)	4629
Military Forces of Turkey	4388
Civilians (Palestine)	3494
Protesters (Iran)	3389
Military Forces of Syria	3287

Lesson Learned

Temporal trend questions represent a challenging routing scenario. Although the query appears analytical and open-ended, answering it requires date filtering, aggregation, and ranking over conflict events. This failure case highlights the need for stronger intent detection between narrative synthesis (GraphRAG) and symbolic reasoning (SPARQL).

Conclusions & Future Work

Project Conclusions

- **Topology reflects reality:** Unsupervised Leiden clustering on the actor co-occurrence graph autonomously recovers real-world conflict theatres, despite never being given country labels.
- **Symbolic (SPARQL) & narrative (GraphRAG) tracks are complementary:** one provides exact, deterministic computation; the other, longitudinal synthesis grounded in retrieved reports — two answers for two different question types, not competing alternatives.

Quantitative Summary

Metric	Score
Modularity / Conductance	0.72 / 0.01
SPARQL Exact Match	95%
GraphRAG Faithfulness	9.30 / 10
GraphRAG Answer Relevance	9.10 / 10

Future Work & Enhancements

- **Hybrid Agentic Routing:** Resolve temporal query bottlenecks by triggering SPARQL aggregations to ground the GraphRAG narrative.
- **Tool Use for SPARQL Generation:** Replace the current rigid generate-then-execute pipeline with an agentic tool use loop.
- **Real-Time Streaming:** Shift from a static historical scope (2015–2023) to continuous, live ACLED REST API streaming to monitor conflict dynamics up to the present day.
- **Semantic IE:** Use LLM information extraction on narrative notes to reveal hidden attributes (weapons, tactical intents).

Disclaimer AI

Use of Artificial Intelligence

AI tools were used as support during the development of this project, specifically for:

- code assistance and debugging;
- brainstorming of implementation strategies;
- reformulation of slide content into academic English.

Additional Personal Experiment

For personal interest, we also developed a graphical interface using Streamlit.

AI was used as coding assistance for the implementation of this interface, which provides an interactive way to query the SPARQL and GraphRAG components of the system.

Key References

- **Leiden Algorithm:** Traag, V. A., Waltman, L., & Van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1), 5233.
- **KG in War Archives:** Yongqin, H., Xushan, C., Anlian, Y., & Shuo, P. (2022). Research on Application of Knowledge Graph in War Archive Based on Big Data. *Int. Conf. on Big Data and Security* (pp. 234-247). Springer.
- **ACLED Semantic Data:** Gastinger, J., Steinert, N., Gründer-Fahrer, S., & Martin, M. (2023). Dynamic Representations of Global Crises: Creation and Analysis of a Temporal Knowledge Graph for Conflicts, Trade and Value Networks. *D2R2*.
- **Text-to-SPARQL QA:** D'Abramo, J., Zugarini, A., & Torroni, P. (2025). Investigating large language models for text-to-sparql generation. *Proc. of the 4th Int. Workshop on Knowledge-Augmented Methods for NLP* (pp. 66-80).
- **LLMs & Conflict Analysis:** Nemkova, A. P., Lingareddy, S. C., Choudhury, S. R., & Albert, M. V. (2025). Do Large Language Models Know Conflict? Investigating Parametric vs. Non-Parametric Knowledge of LLMs for Conflict Forecasting. *arXiv:2505.09852*.
- **Hybrid LLM Routing:** Abedini, Y., & Heydarnoori, A. (2025). Hybrid LLM Routing for Efficient App Feedback Classification. *arXiv:2507.08250*.
- **RAGAS Evaluation:** Es, S., James, J., Anke, L. E., & Schockaert, S. (2024). Ragas: Automated evaluation of retrieval augmented generation. *Proc. of the 18th EACL: System Demonstrations* (pp. 150-158).